

# Bundesliga Player Valuation System

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Soccer Data Analytics Hackathon

# Prompt 2: Build a Transparent Player Valuation Metric

## Prompt

Define an interpretable attacking or defensive metric using event data to compare players within the 2023/24 Bundesliga season.

### Key Requirements:

- Transparent methodology
- Clear metric definition
- Player leaderboard
- Case study analysis

## Approach

Multi-dimensional rating system with 8 granular metrics across attack, defense, and passing dimensions.

### Key Features:

- Position-aware weighting
- Transparent event scoring
- Per-90 normalization for fair comparison
- Players rated from 962,990 events

## Results

Going beyond traditional stats

### What's New:

- Multi-dimensional analysis (not just goals + assists)
- Role-specific evaluation (defenders aren't penalized for not scoring)
- Event-level granularity (captures 962,990 actions)
  - Transparent methodology (no "black box")

# Our Rating System: Granular Metrics Across 3 Dimensions

## 1. Three Core Dimensions

- Attack: Finishing, Chance Creation, Dribbling
- Defense: Ball Winning, Defensive Actions
- Passing: Accuracy, Ball Progression, Long Distribution

## 2. Transparent Event Weighting

- Every action has a clear point value
- Example: Goal = 10.0 pts, Assist = 8.0 pts, Duel Won = 3.0 pts

## 3. Position-Aware Scoring

- Forwards weighted 50% Attack, 30% Passing, 20% Defense
- Midfielders weighted 33% Attack, 34% Passing, 33% Defense
- Defenders weighted 20% Attack, 30% Passing, 50% Defense

## 4. Fairness Through Normalization

- Per-90 minutes adjustment (accounts for playing time)
- Z-score standardization within positions (0-100 scale)

## 5. Captures "Invisible" Contributions

- Progressive passes, ball winning, defensive work
- Goes beyond goals + assists

# Approach

## 1. Hierarchical Structure

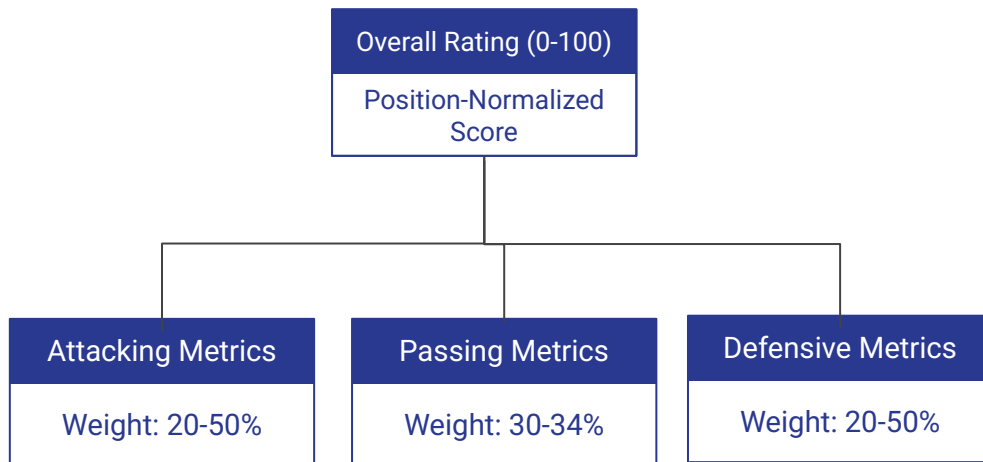
Overall rating derived from 3 dimensions, each composed of 2-3 granular metrics. Clear top-down flow ensures transparency.

## 2. Position-Adaptive Weighting

Same metrics, different importance by role. Forwards prioritize attack (50%), defenders prioritize defense (50%).

## 3. Interpretable by Design

Every rating traces back to specific events. No "black box" - stakeholders can audit and explain any score.



# Deliverables

## Project Outputs:

- **GitHub Repository: Full code + documentation**
- **Player ratings dataset (Parquet & CSV)**
- **Reproducible analysis pipeline**
- **Comprehensive methodology documentation**

## Future Enhancements:

- **Integrate xG and expected threat models**
- **Add game context (score, opponent quality)**
- **Multi-season tracking and development curves**
- **Market value correlation analysis**

# Thank You!

## Questions & Discussion

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**GitHub: <https://github.com/Tashbhilare/SoccerImpectHackathon>**